

Database Systems		
Lab #:	02	
Lab Title:	Database keys, Format, Caption, Input masks and Validation rules.	
Submitted by:		
Names		Registration #
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Rubrics name & number					Marks	
					In-Lab	Post-Lab
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.					
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.					
Design	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.					
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.					
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.					
	R13: Management of Team Work: Ability to appreciate, understand and work with multidisciplinary team members.					
Rubrics #	R2	R6	R8	R9	R12	R13
In Lab						
Post- Lab						

Lab No: 02

Title: Database keys, Format, Caption, Input masks and Validation rules.

Objective:

Keys are used to establish and identify relationships between tables and also to uniquely identify any record or row of data inside a table. A Key can be a single attribute or a group of attributes. Format adjusts how the data are displayed as output. For Date/Time data types, this will adjust the date and time format used. The Input Mask Wizard only works with text or date field type. Input Mask creates a predefined structure into which the data for this field must be entered. Validation rules stop users from entering incorrect data into your tables.

In Lab Task 1:

The SIN consists of nine digits. Each group has three digits separated by the dash ('-') character. The SIN MUST have exactly nine digits. (e.g. 435-542-087).

Design View:

Field Name	Data Type
ID	Number
SIN	Short Text

General	
Field Size	255
Format	
Input Mask	
Caption	
Default Value	
Validation Rule	Like "###-###-###"
Validation Text	
Required	No
Allow Zero Length	Yes
Indexed	No
Unicode Compression	Yes
IME Mode	No Control
IME Sentence Mode	None
Text Align	General

Datasheet View:

Views Clipboard Sort & Filter		
All Acc... <<		
Search...		
Tables >>		
SIN		
	ID	SIN
	340	345-976-098
	230	678-089-978
	430	709-787-767
	290	374-234-345
	630	246-375-754
	860	267-874-483
	760	247-376-766
*	0	

Validation rules check:

All Acc... <<		
Search...		
Tables >>		
SIN		
	ID	SIN
	340	345-976-098
	230	678-089-978
	430	709-787-767
	290	374-234-345
	630	246-375-754
	860	267-874-483
	760	247-3876-766
*	0	

Microsoft Access

OK Help

In lab Task 2:

Add an input mask to the TelNum field that allows entry of numbers like: (403) 789-1234. The area code should be optional. (Hint: don't forget to change the data type of TelNum first. Also, consider using #).

Design View:

General Lookup	
Field Size	255
Format	
Input Mask	* (###) *000\ -0000
Caption	telephone number
Default Value	
Validation Rule	
Validation Text	
Required	No
Allow Zero Length	Yes
Indexed	Yes (Duplicates OK)
Unicode Compression	Yes
IME Mode	No Control
IME Sentence Mode	None
Smart Tags	

Datasheet View:

Field Name	Data Type
Tel Num	Text
POST CODE	Text
SALARY	Number
date of birth	Date/Time
date joined	Date/Time

Input Mask check:

	telephone number
	(301) 559-0502
*	() -

In Lab Task 3:

Enter an input mask for the postal code so that it restricts the format to three alphanumeric characters, followed by a space, followed by three other alphanumeric characters (for instance, T2N 1N4).

Design View:

All Accounts

Search...

Tables

Postal Code

Postal Code

Field Name	Data Type
ID	AutoNumber
Postal code	Short Text
Address	Short Text
Country	Short Text

General

Lookup

Field Size	255
Format	
Input Mask	>LOL\ OLO"
Caption	
Default Value	
Validation Rule	
Validation Text	
Required	No
Allow Zero Length	Yes
Indexed	Yes (Duplicates OK)
Unicode Compression	Yes
IME Mode	No Control
IME Sentence Mode	None
Text Align	General

Datasheet View:

All Acc...					
Postal Code					
ID	Postal code	Address	Country	Click to Add	
1	D2E 6H5	Lahore	Pakistan		
2	G6H 5J7	California	Canada		
3	J6F 8K9	Naqoyo	Japan		
4	T5H 3K6	Beijing	China		
5	U6G 7M8	Azad Kashmir	Pakistan		
6	Y7I 5E8	Nottingham	United Kingdom		
7	H3O 2X6	Birmingham	Landon		
8	G4Y 6T5	Rawalpindi	Pakistan		
*	(New)				

In lab Task 4:

Set validation rules (and respective validation texts) for the DOB and DateJoined fields so that they are always before or equal to today's date.

Design View:

All Acc...		
people		
Field Name	Data Type	
Customer ID	AutoNumber	
First Name	Short Text	
Last Name	Short Text	
Address	Short Text	
city	Short Text	
State	Short Text	
Birthdate	Date/Time	
Zip	Number	
Today Date	Date/Time	
Joined Date	Date/Time	

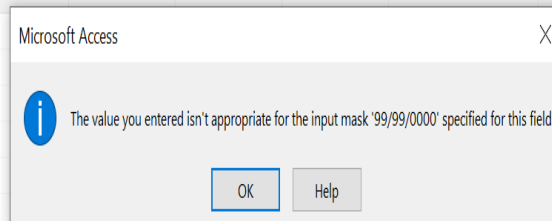
General	Lookup
Format	
Input Mask	99/99/0000
Caption	"DOB"
Default Value	
Validation Rule	<Date()
Validation Text	Invalid Date
Required	No
Indexed	No
IME Mode	No Control
IME Sentence Mode	None
Text Align	General
Show Date Picker	For dates

Datasheet View:

Customer ID	First Name	Last Name	Address	city	State	"DOB"	Zip	Today Date	Joined Date	Click to Add
1	Sarah Zee	Rost	123 Main street	Bhakkar	FL	1/1/2022	33993	7/27/2023	1/15/2012	
2	Jiya	Smith	333 park avenue	5 DI Khan	NY	1/2/2090	14222	4/14/2024	3/5/2023	
3	Hannah	KirK	101 smith street	Kashmir	IA	1/3/2004	52327	12/23/2021	9/12/2020	
4	Zoe William	Riker	8 Right ave no 4	Faislabad	AK	1/4/2021	999801	6/23/2025	12/3/1993	
5	Malik Budhan	Williams	22 pea street 0	Karachi	NY	1/5/2020	14223	10/31/2023	4/6/2003	
*	(New)						0		__/__/__	

Validation rule and Text check:

Customer ID	First Name	Last Name	Address	city	State	"DOB"	Zip	Today Date	Joined Date	Click to Add
1	Sarah Zee	Rost	123 Main street	Bhakkar	FL	1/1/2022	33993	7/27/2023	1/15/2012	
2	Jiya	Smith	333 park avenue	5 DI Khan	NY	1/2/2090	14222	4/14/2024	3/5/2023	
3	Hannah	KirK	101 smith street	Kashmir	IA	1/3/2004	52327	12/23/2021	9/12/2020	
4	Zoe William	Riker	8 Right ave no 4	Faislabad	AK	1/4/2021	999801	6/23/2025	12/3/1993	
5	Malik Budhan	Williams	22 pea street 0	Karachi	NY	1/5/2020	14223	10/31/2023	__/0/878	
*	(New)						0			



In lab Task 5:

Set a validation rule and text for the salary field so that it is greater than zero.

Design View:

All Acc...

Search...

Tables

Lab2 Inlab#05

Lab2 Inlab#05

Field Name	Data Type
Employee Name	Short Text
Status	Short Text
Contact No	Short Text
Age	Number
Salary	Number

Field

General

Lookup

Field Size	Long Integer
Format	
Decimal Places	Auto
Input Mask	
Caption	
Default Value	0
Validation Rule	> 0
Validation Text	Salary must be greater than zero
Required	No
Indexed	No
Text Align	General

Datasheet View:

Employee Name	Status	Contact No	Age	Salary	Click to Add
Malik Noran	Staff	3234556779	25	40000	
Bilal Khan	Assistant	3544567789	28	50000	
Zoe Rohan	Manager	3456798753	40	90000	
Sarah	Asst. Manager	3445567896	35	6000	
Hannah	Marketing Manager	3563829904	22	34780	
Asghar zee	Data Analyst	3876655556	30	43000	
Kamran Hassan	Graphic Designer	3679008877	37	50000	
Farhan Farman	Software Engineer	3976656554	26	43000	
*			0	0	

Validation rules check:

All Access Objects

Search...

Tables

Lab2 Inlab#05

Employee Name	Status	Contact No	Age	Salary	Click to Add
Malik Noran	Staff	3234556779	25	40000	
Bilal Khan	Assistant	3544567789	28	0000	
Zoe Rohan	Manager	3456798753	40	90000	
Sarah	Asst. Manager	3445567896	35	6000	
Hannah	Marketing Manager	3563829904	22	34780	
Asghar zee	Data Analyst	3876655556	30	43000	
Kamran Hassan	Graphic Designer	3679008877			
Farhan Farman	Software Engineer	3976656554			

*

Microsoft Access

! Salary must be greater than zero

OK Help

Post lab Tasks:

Post lab Task 1:

Fill in at least one record in your table. Try to enter incorrect data into some fields with validation rules to see how your validation text is displayed. Also, notice that DOB is displayed as Birth Date because of the caption that was set.

Design View:

All Access

Search...

Tables

Forms

people

peopleform

Field Name	Data Type
Customer ID	AutoNumber
First Name	Short Text
Last Name	Short Text
Address	Short Text
city	Short Text
State	Short Text
Birthdate	Date/Time
Zip	Number
Creditlimit	Number

General

Lookup

Field Size	Long Integer
Format	
Decimal Places	Auto
Input Mask	
Caption	
Default Value	0
Validation Rule	<= 2000
Validation Text	THE Credit limit must be below 2000
Required	No
Indexed	No
Text Align	General

Datasheet View:

Clipboard

people

Sort & Filter

Records

Find

Text Formatting

Customer ID	First Name	Last Name	Address	city	State	"Birth Date"	Zip	Creditlimit
1	Sarah Zee	Rost	123 Main street 4	Bhakkar	FL	1/1/2022	33993	1670
2	Jiya	Smith	333 park aven5	DI Khan	NY	1/2/2090	14222	1200
3	Hannah	KirK	101 smith street2	Kashmir	IA	1/3/2004	52327	0
4	Zoe William	Riker	8 Right ave no 4	Faislabad	AK	1/4/2030	999801	0
5	Malik Budhan	Williams	22 pea street 01	Karachi	NY	1/5/2020	14223	1400
*	(New)						0	0

Form:

All Acc... << people peopleform

Search...

Tables

people

Forms

peopleform

people

Customer ID 1

First Name Sarah Zee

Last Name Rost

Address 123 Main street 4

city Bhakkar

State FL

Birthdate 1/1/2022

Zip 33993

Creditlimit 1670

Validation rules check:

All Acc... << people

Search...

Tables

people

Forms

peopleform

Customer ID	First Name	Last Name	Address	city	State	"Birth Date"	Zip	Creditlimit
1	Sarah Zee	Rost	123 Main street 4	Bhakkar	FL	1/1/2022	33993	1000
2	Jiya	Smith	333 park aven5	DI Khan	NY	1/2/2090	14222	200
3	Hannah	KirK	101 smith street2	Kashmir	IA	1/3/2004	52327	0
4	Zoe William	Riker	8 Right ave no 4	Faislabad	AK	1/4/2030	999801	0
5	Malik Budhan	Williams	22 pea street 01	Karachi	NY	1/5/2020	14223	7600
*	(New)						0	0

Microsoft Access

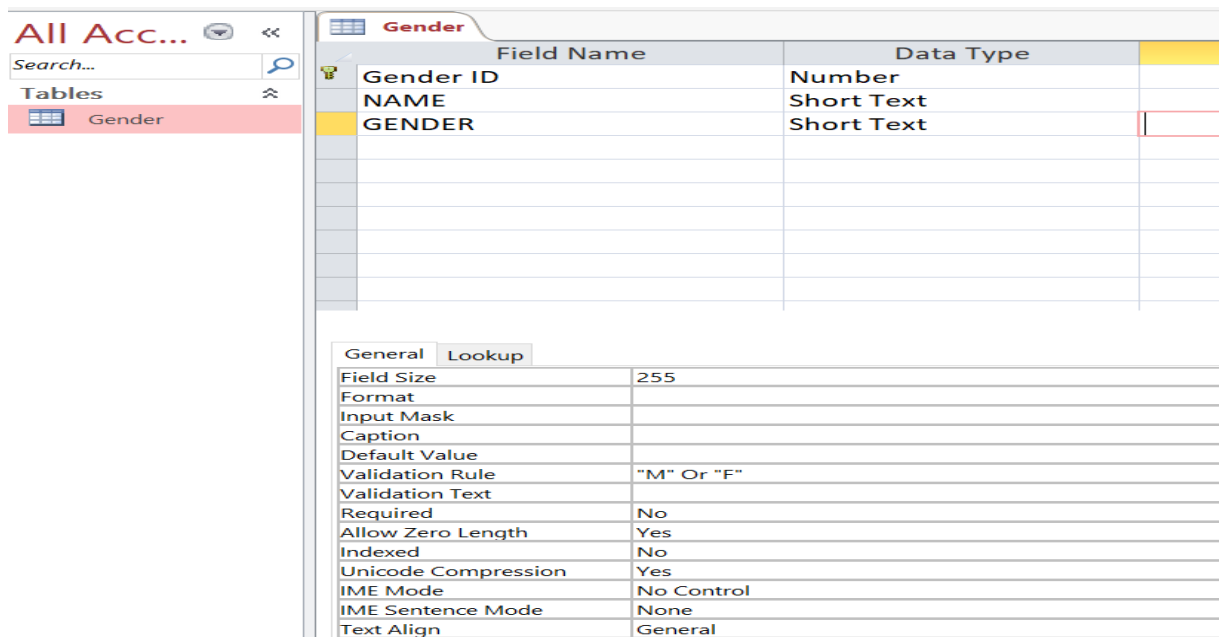
THE Credit limit must be below 2000

OK Help

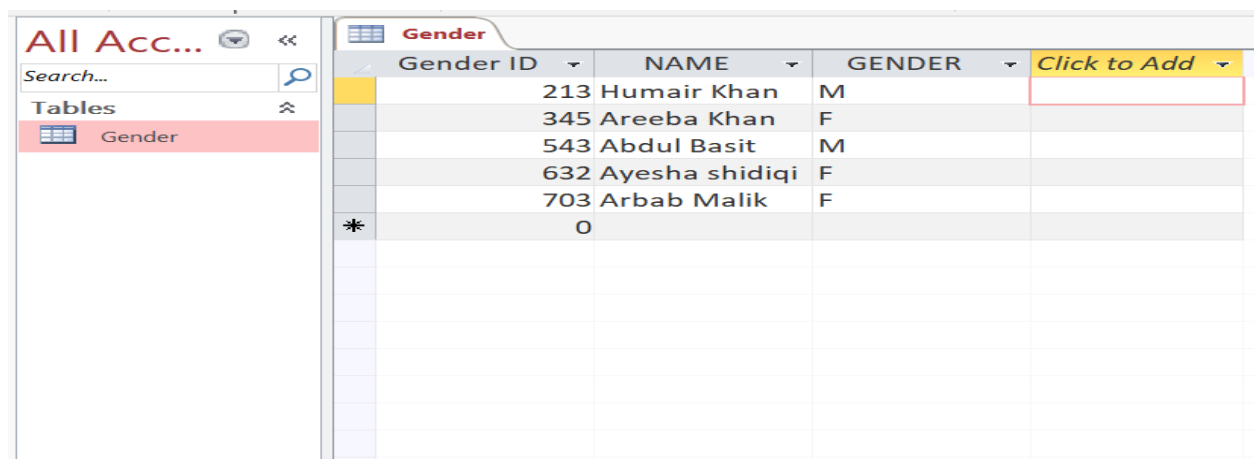
Post lab Task 2:

Create a validation rule so that the Gender field is limited to either "M" or "F" (Male or Female). The field can also be left blank.

Design View:



Datasheet View:



Post lab Task 3:

Create a validation rule so that the Email field is restricted to only this format. (xyz@ciit-attack.edu.pk) .

Design View:

All Acc... << >>

Search...

Tables

Gender

Gender

Field Name	Data Type
Gender ID	Number
NAME	Short Text
GENDER	Short Text

Field

General Lookup

Field Size	255
Format	
Input Mask	
Caption	
Default Value	
Validation Rule	"M" Or "F"
Validation Text	
Required	No
Allow Zero Length	Yes
Indexed	No
Unicode Compression	Yes
IME Mode	No Control
IME Sentence Mode	None
Text Align	General

Datasheet View:

Views Clipboard Sort & Filter Records Find

All Acc... << >>

Search...

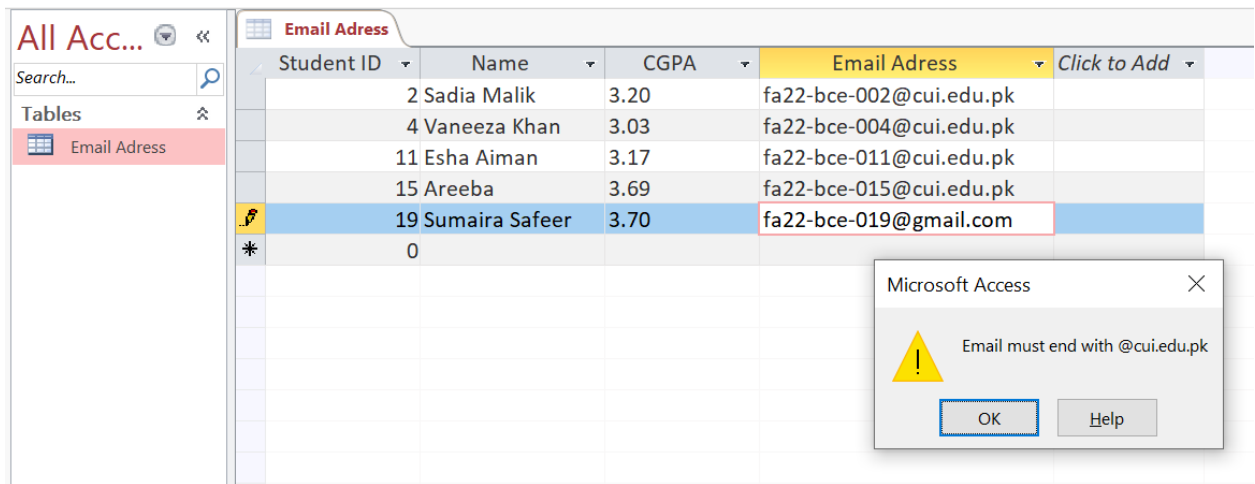
Tables

Email Adress

Email Adress

Student ID	Name	CGPA	Email Adress	Click to Add
2	Sadia Malik	3.20	fa22-bce-002@cui.edu.pk	
4	Vaneeza Khan	3.03	fa22-bce-004@cui.edu.pk	
11	Esha Aiman	3.17	fa22-bce-011@cui.edu.pk	
15	Areeba	3.69	fa22-bce-015@cui.edu.pk	
19	Sumaira Safeer	3.70	fa22-bce-019@cui.edu.pk	
*	0			

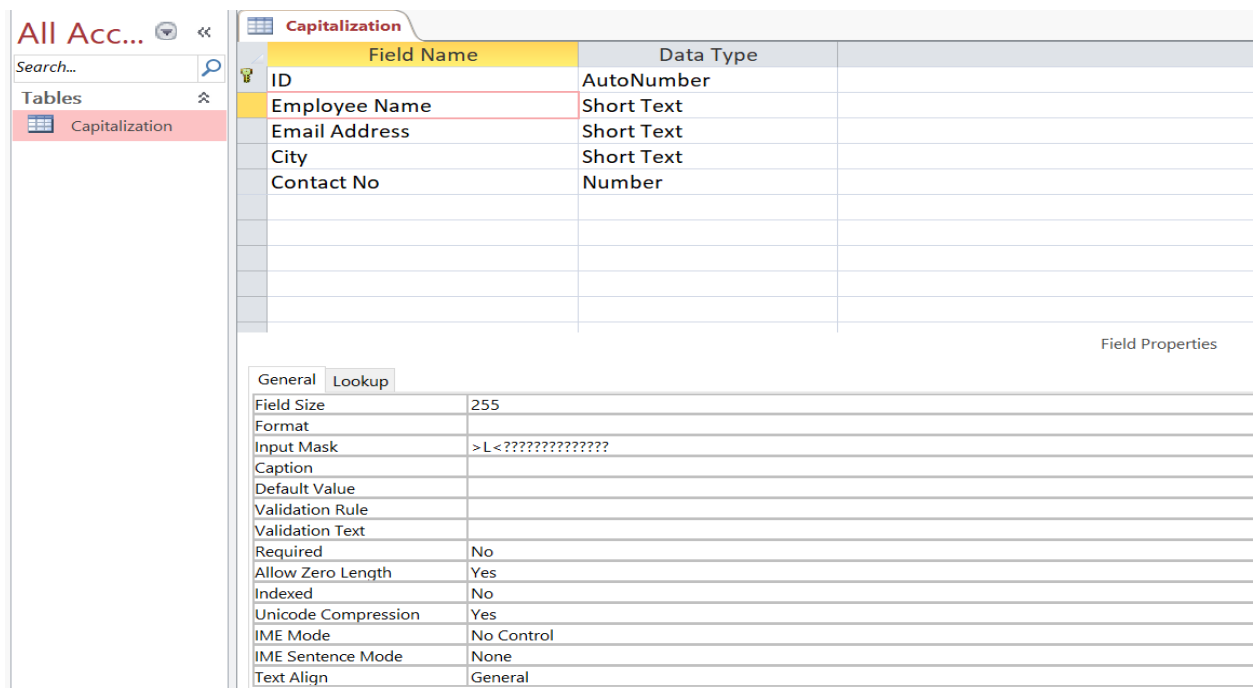
Validation rules check:



Post lab task 4:

A first or last name with the first letter automatically capitalized. Add an input mask for this data. e.g Herry Potter

Design View:



Datasheet View:

All Acc...

Search...

Tables

Capitalization

Capitalization

ID	Employee Name	Email Address	City	Contact No	Click to Add
1	Sumaira	@ciit-attock.edu.pk	Mianwali	346382672	
2	Aaan zarsham	@ciit-attock.edu.pk	Islamabad	345178888	
3	Farhan khan	@ciit-attock.edu.pk	Karachi	345345670	
4	Saqiob hassan	@ciit-attock.edu.pk	Lahore	375679900	
5	Tayyaba amjid	@ciit-attock.edu.pk	Dera Ismail	376526680	
6	Hannah zoe	@ciit-attock.edu.pk	Abbtabad	362819291	
7	Hamza khan	@ciit-attock.edu.pk	Chistian	387897650	
*	(New)			0	